PSP Prod IBOB Aurora PostgreSQL migration

Current database object layout

	OWNER	OBJECT_TYPE	OBJECT_COUNT
1	IBOBADM	INDEX	22
2	IBOBADM	INDEX PARTITION	978
3	IBOBADM	INDEX SUBPARTITION	6846
4	IBOBADM	LOB	4
5	IBOBADM	LOB PARTITION	652
6	IBOBADM	LOB SUBPARTITION	4564
7	IBOBADM	TABLE	10
8	IBOBADM	TABLE PARTITION	163
9	IBOBADM	TABLE SUBPARTITION	1141

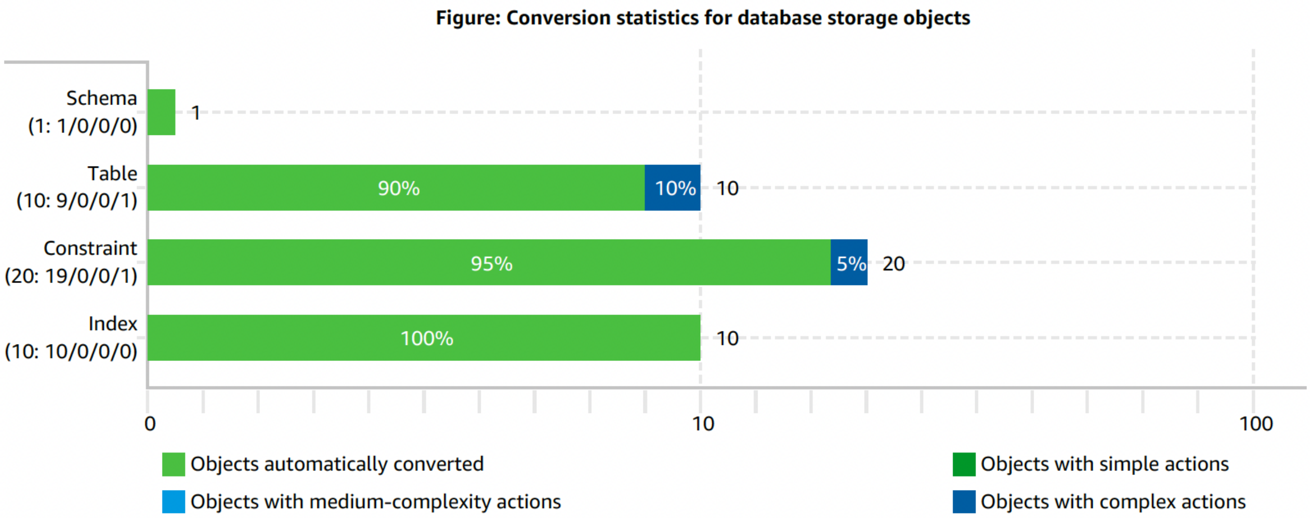
Object Sizes

Table	Table Size (GB)	Index Size (GB)	CLOB Size (GB)	Total Size (GB)
PSP_SOURCE_SYSTEM_TRANSMISSION	2125	283	1714	4122
PSP_SAP_METHOD_CALL	455	211		666
PSP_QBDT_REQUEST_INFO	74	30		104
PSP_HCM401K_COMPANY_POLICY	0	0	0	0
PSP_HCM401K_COMPANY_QBDT_PITEM	0	0	0	0
PSP_HCM401K_EMPLOYEE_DEDUCTION	0	0	0	0
PSP_HCM401K_POLICY	0	0	0	0
Total	2654	524	1714	4892

Assessment report

Summary

IBOB



No statistics available for database code objects for IBOB/Secondary database.

Full Report

psp_prod_ibob_sct_migration_assessment_apg.pdf

Aurora Postgres Database

Setup

We have used IPS to provision aurora postgre database cluster.

Source code: git code

Configuration

Category	Config Name	Value	Comments
Database	Postgres Version	14.3	
	timezone	US/Pacific	
Database Security	Authentication	DB password login	create db user with password
	Authorization	Grant respective roles/privilges to users	User can access objects based on role/privileges assigned
Data Encryption	Storage Encryption	enabled	Encryption at rest
Network security	Security Groups	SG/CIDR	To restrict connectivity to database
	rds.force_ssl	1	To enable secure TCP connection. ssl_min_protocol_version is set to TLSv1.2

Best Practice DB Parameters	**Vaccum :** autovacuum rds.force_autovacuum_logging_level log_autovacuum_min_duration **Performance** rds.enable_plan_management apg_plan_mgmt.capture_plan_baselines apg_plan_mgmt.use_plan_baselines auto_explain.log_analyze auto_explain.log_buffers auto_explain.log_min_duration auto_explain.log_timing auto_explain.log_nested_statements auto_explain.log_triggers default_statistics_target **Maintenance** max_parallel_maintenance_workers max_parallel_workers max_parallel_workers_per_gather random_page_cost work_mem **Logging** log_min_messages log_min_error_statement log_filename log_lock_waits log_min_duration_statement log_statement log_temp_files	on log 0 1 automatic true 1 1 200 1 1 1 512 64 64 4 1 65536 ERROR INFO postgresql.log.%Y-%m-%d 1 1000 ddl 65536	

Oracle Vs Aurora Postgres DB Parameters - Draft

Oracle Parameter	Oracle	Aurora Postgres Parameter	Aurora Postgres	Comments
timezone	US/Pacific	timezone	US/Pacific	
processes	6000	max_connections	6000	
NLS_CHARACTERSET	AL32UTF8	Encoding	UTF8	

List of Supported Extensions in Postgres 13.4

```
address_standardizer
address_standardizer_data_us
amcheck
apg_plan_mgmt
aurora_stat_utils
autoinc (contrib-spi)
aws_commons
aws_lambda
aws_ml
aws_s3
bloom
bool_plperl
btree_gin
btree_gist
citext
cube
dblink
dict_int
dict_xsyn
earthdistance
fuzzystrmatch
hll
hstore
hstore_plperl
insert_username (contrib-spi)
intagg
intarray
ip4r
isn
jsonb_plperl
log_fdw
ltree
moddatetime (contrib-spi)
oracle_fdw
orafce
pg_bigm
pg_buffercache
pg_cron
pg_freespacemap
pg_hint_plan
pg_partman
pg_prewarm
pg_proctab
pg_repack
pg_similarity
pg_stat_statements
pg_trgm
pg_visibility
pgAudit
pgcrypto
pglogical
pglogical_origin
pgrouting
pgrowlocks
pgstattuple
pgtap
plcoffee
plls
plperl
plpgsql
plprofiler
pltcl
plv8
PostGIS
postgis_raster
postgis_tiger_geocoder
postgis_topology
```

```
postgres_fdw
prefix
rdkit
rds_activity_stream
rds_tools
refint (contrib-spi)
sslinfo
tablefunc
tds_fdw
test_parser
tsm_system_rows
tsm_system_time
unaccent
uuid-ossp
wal2json
```

Schema Conversion

Data Type Mapping

Criteria followed for target datatype w.r.t source NUMER datatype

Precision(m)	Scale(n)	Oracle	PostgreSQL	Comments
<5	0	NUMBER(m,n)	SMALLINT	Documentation Page
5 >= m <= 9	0	NUMBER(m,n)	INT	
10<= m <=18	0	NUMBER(m,n)	BIGINT	
m+n <= 15	n>0	NUMBER(m,n)	DOUBLE PRECISION	
m+n > 15	n>0	NUMBER(m,n)	NUMERIC	

	Oracle	Postgres
1	CLOB	TEXT
2	NUMBER(1)	SMALLINT
3	NUMBER(10)	BIGINT
4	NUMBER(19)	NUMERIC(19,0)
5	NUMBER(19) NOT NULL	NUMERIC(19,0) NOT NULL
6	NUMBER(19,7)	NUMERIC(19,7)
7	TIMESTAMP(6)	timestamp(6) without time zone
8	TIMESTAMP(6) NOT NULL	timestamp(6) without time zone NOT NULL
9	VARCHAR2(100 Char)	character varying(100)
10	VARCHAR2(20 Char)	character varying(20)
11	VARCHAR2(200 Char)	character varying(200)
12	VARCHAR2(255 Char)	character varying(255)
13	VARCHAR2(255 Char) NOT NULL	character varying(255) NOT NULL
14	VARCHAR2(256 Char)	character varying(256)
15	VARCHAR2(30 Char)	character varying(30)
16	VARCHAR2(40 Char)	character varying(40)
17	VARCHAR2(400 Char)	character varying(400)
18	VARCHAR2(4000 Char)	character varying(4000)

DataType Mapping Analysis: Based on actual data store , captured analysis for target data type at https://docs.google.com/spreadsheets/d/1T2j9p8C8pQpkVwY9eG0C6x7pH3vBiVZi64gJ0Jc6L-c/edit#gid=1772497532

Schema DDLs

Schema DDLs code

Partitioning

PSP IBOB database has table (SOURCE_SYSTEM_TRANSMISSION) in Oracle. This table is partitioned (RANGE-LIST). As part of Oracle Exit journey, we need to migrated this to Aurora PostgreSQL, where both database engines has different partitioning aspect/limitations/features. This document addresses those limitation, mitigations and decision on partitioning for this table.

Table	Table Size (GB)	Index Size (GB)	CLOB Size (GB)	Total Size (GB)
PSP_SOURCE_SYSTEM_TRANSMISSION	1990	254	1523	3767

There are below limitations on table partitioning in PostgreSQL :

1. Partitioning key has to be part of primary key on partitioned table. This limitation exists in latest version (Postgres **14**) as well. We are currently on **13.4** PostgreSQL (AWS Aurora).
2. On partitioned table, Scope of Primary key constraint is restricted to partitions not global.

	Primary Key (Oracle)	Partitioning Key (Oracle)	Primary Key (PostgreSQL)	Partitioning Key (PostgreSQL)	Partition Type
1	SOURCE_SYSTEM_TRANSMISSION_SEQ, REALM_ID	CREATED_DATE	SOURCE_SYSTEM_TRANSMISSION_SEQ, REALM_ID	CREATED_DATE	Range (monthly) - List

Keeping above limitations in mind, we are going ahead with same partitioning as Oracle for below reasons:

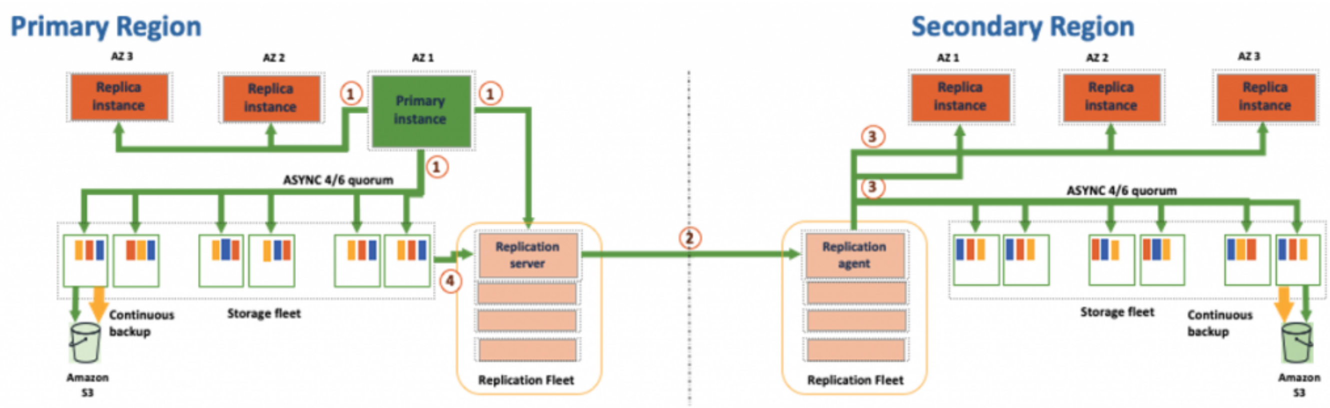
- IBOB database and its tables use case. It stores data which is most of audit purpose/nature. There are no deletes in any of these tables.
- It would help in optimising data validation of latest/data based on time range.
- It also supports scope of primary key constraints at partition level. In addition to this, we also have ensured way to restrict primary keys via GUID (java code) to insert unique keys across table and Postgres will ensure uniqueness at partition level.

- How are we mitigating limitations ?
  - Data Integrity
    - Postgres supports scope of primary key constraints at partition level. And we have ensured way to restrict primary keys via GUID (java code) to insert unique keys across table and Postgres will ensure uniqueness at partition level.
  - Performance
    - Since without partitioning key, query would go for full table scan to get data. We would need to add partition key (CREATED_DATE) as part of where clause for SELECTs and UPDATEs. More details on performance testing are shown below:

- Why we are following this criteria and add more details on partitioning details
  - Based on data access pattern from application.
  - It would help in optimising data validation of latest/data based on time range.
- What are we loosing when we are not adding partitioning key as part of primary key.
  - We will not have global index and unique constraint defined at table level (across all partitions).

Aurora PostgreSQL HA & DR

- Aurora Global database for DR. Enable multi-AZ for HA in same region and enable Aurora Global for cross-region.



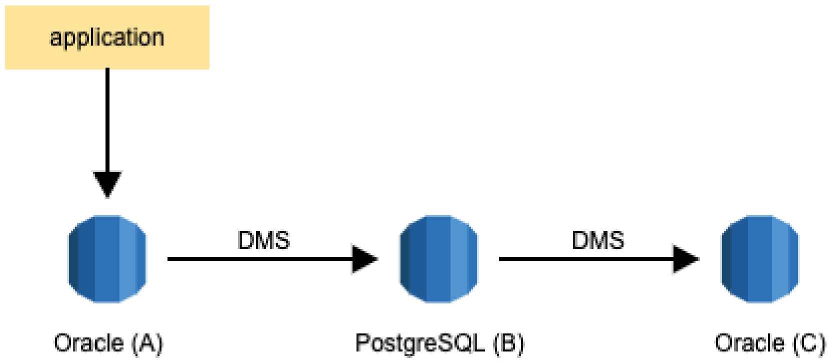
Data Replication

Design principles

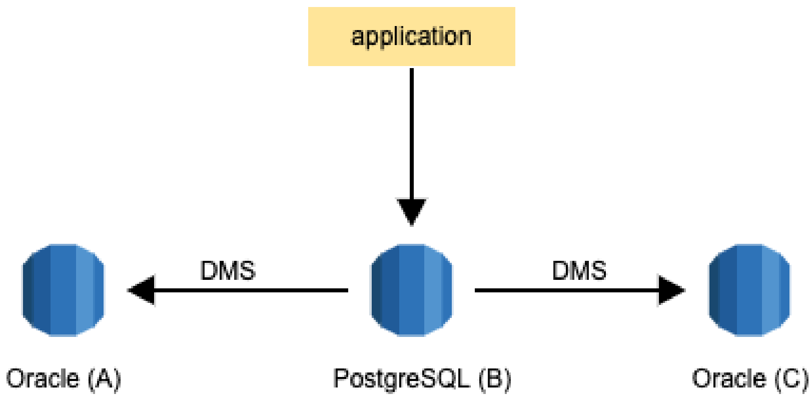
- Maintain CDC replication for latest 3 months data for SST table
- one time migration of historical data for SST table
- Break down each year data and replicate respectively (to achieve performance)
- Maintain less number of tasks specially for CDC replication (to get clear visibility on data to handle and maintainability for replication)

```
SQL> SELECT    /*+ PARALLEL(8) */
  2    modified_date - created_date, SOURCE_SYSTEM_TRANSMISSION_SEQ
   3  FROM IBOBADM.PSP_SOURCE_SYSTEM_TRANSMISSION sst
   4  where created_date > sysdate - 90
   5  and modified_date - created_date > interval '2' day;

no rows selected

SQL> SELECT    /*+ PARALLEL(8) */
   2    modified_date - created_date, SOURCE_SYSTEM_TRANSMISSION_SEQ
   3  FROM IBOBADM.PSP_SOURCE_SYSTEM_TRANSMISSION sst
   4  where created_date > sysdate - 365
   5  and modified_date - created_date > interval '2' day;

no rows selected
```

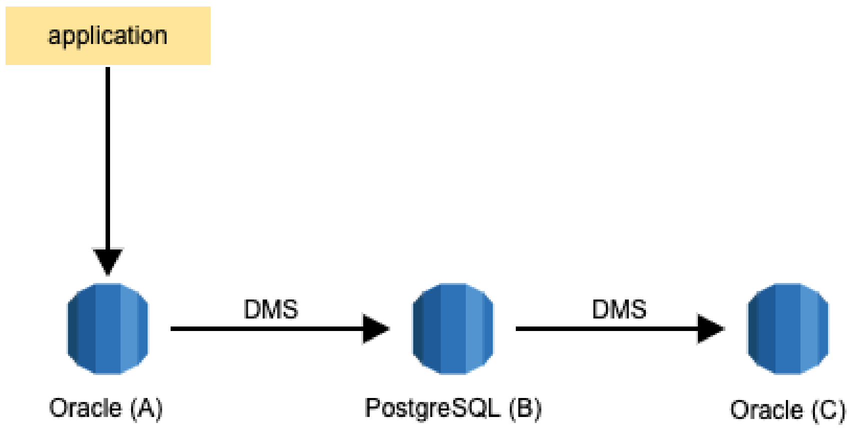
Replication Setup



At cutover, the application points to target PostgreSQL (B) and the replication stream from Oracle (A) to PostgreSQL (B) is continued. The replication stream from PostgreSQL (B) to Oracle(C) continues to run in case rollback is required.



To perform a rollback, the applications start pointing to database Oracle (C), that includes the changes that occurred on PostgreSQL database (B).



Cutover & Rollback Strategy

Cutover steps

- Stop all application on the Oracle database.
- Lock account on Oracle database
- Let the AWS DMS task replicate remaining changes to the target PostgreSQL database.
- Unlock accounts on PostgreSQL database.
- Start all applications back on the source Oracle database

Rollback steps

- Stop all application on the target PostgreSQL database.
- Lock accounts on PostgreSQL database.
- Let the AWS DMS task replicate remaining changes back to the source Oracle database.
- Unlock accounts on Oracle database
- Start all applications back on the source Oracle database.

Replication Configuration

Oracle to Aurora Postgres

	Replication type	Replication scope	Duration (hh:mm)	total_rowcount	Data_Size(GB)
1	Historical + CDC	SMC	08:46	1,403,836,988	
		QBRI	01:43	194,129,509	
2	Historical load	2009 year data	00:06	4,131,121	9
3	Historical load	2010 year data	00:43	10,171,482	25
4	Historical load	2011 year data	01:10	16,010,138	34
5	Historical load	2012 year data	01:30	22,050,518	47
6	Historical load	2013 year data	02:03	25,232,805	66
7	Historical load	2014 year data	05:00	34,646,398	89
8	Historical load	2015 year data	10:00	112,575,822	258
9	Historical load	2016 year data	09:45	127,091,939	291
10	Historical load	2017 year data	08:43	130,216,174	296

11	Historical load	2018 year data	08:58	123,364,300	284
12	Historical load	2019 year data	06:57	109,407,270	237
13	Historical load	2020 year data	05:59	112,276,695	237
14	Historical load	2021 year data	10:20	115,359,282	243
15	Historical + CDC	2022 + CDC	04:55	23,967,322	

SST table

Current State

- Detached partitions as separate tables for each Year data

Target state for cutover

- Day before testing/cutover, Attach all partitions back

Aurora Postgres to Oracle

1. Create a Oracle RDS instance
2. Configure source and target database
3. Schema deployment on target database
4. Specify source and target endpoints using AWS DMS console
5. Create a task and migrate data
6. Monitor Full load + CDC replication using replication metrics

	Replication type	Replication scope	Duration (hh:mm)
1	Historical + CDC	All tables	99:50

Reference : DMS replication code

Testing strategy

Use case: Read & Write workflow testing (One time copy/latest data/refresh)

- Clone existing database to another cluster
- Truncate/drop all 3 tables (SMC,QBRI, SST with latest 3 months partitions)
- Recreate all 3 tables (SMC,QBRI, SST with latest 3 months partitions)
- task 1 -> replicate non-partition / all tables except SST (ETA time: ~12 hrs)
- task 2 -> replicate table partitions from Jan2022 (ETA time: ~7 hrs)
- once replication completes, we are ready for testing (mostly next day)
- On the day of cutover/testing, Attach all historical partitions to SST table
    - At this time, replication is not working from oracle -> postgres

Note : As per discussion on 22 Mar 2022 continuous replication is not required for any workflow testing. All of above steps would be replaced by refresh automation.

Detailed status

Task	Sub-Task	Status	Details	Owner	Comments
DB Setup and Configuration	• Prep-work • Manual schema conversion • DB Setup	COMPLETED		Mayank	

Aurora Postgres Database	• limitations • Partitioning • AWR on Postgres		• PSP_QBDT_REQUEST_INFO_FK1 on PSP_QBDT_REQUEST_INFO will not exist because of absence global scope of PK on PSP_SOURCE_SYSTEM_TRANSMISSION table • Test partitioning automation		
Data Replication	Oracle Aurora Postgres	COMPLETED	• Replication performance • Replication monitoring		
	Aurora Postgres Oracle (A B C)	COMPLETED	• Test loopback prevention settings • Replication performance • Replication monitoring		
DB Performance	DB Server Performance	COMPLETED			
	SELECT Queries Performance	COMPLETED			
DB Monitoring	• DB Monitoring Dashboards • Alerts & Pager	COMPLETED			
HA / DR		IN PROGRESS			
Data to Downstream	• IAC data ingestion	IN PROGRESS			
Tip-Toe & Switchover	• Switchover testing • Switchback testing	COMPLETED			